Machine Learning

Lecture 1

Introduction to Machine Learning

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Lecture Objectives

Understand what Machine Learning is

Identify real-world applications

Understand basic terminology

What is Machine Learning?

Machine learning is a discipline that deals with programming the systems to make them automatically learn and improve with experience. Here, learning implies recognizing and understanding the input data and taking informed decisions based on the supplied data.

Simply machine learning is:

- Field of AI that enables systems to learn from data
- Improves performance without explicit programming
- Learns patterns and relationships

Traditional Programming vs ML

Traditional: Rules + Data → Output

ML: Data + Output → Rules (Model)

Focus on learning from data

What we want to do

- Algorithms that can improve their performance using **training data**
- Typically the algorithm has a (large) number of parameters whose values are learnt from the data
- Can be applied in situations where it is very challenging (= impossible) to define rules by hand, e.g.:
  - Facedetection
  - Speechrecognition
  - Stockprediction

Why Machine Learning?

- Handles large data
- Adapts to new patterns
- Reduces manual effort
- Improves decision making
- Automation
- Accuracy improvement
- Scalability

Purpose of Machine Learning

The purpose of machine learning is not building an automated duplication of intelligent behavior but using the power of computers to complement and supplement human intelligence.

For example, machine learning programs can scan and process huge databases detecting patterns that are beyond the scope of human perception.

Applications of Machine Learning

The developed machine learning algorithms are used in various applications such as:

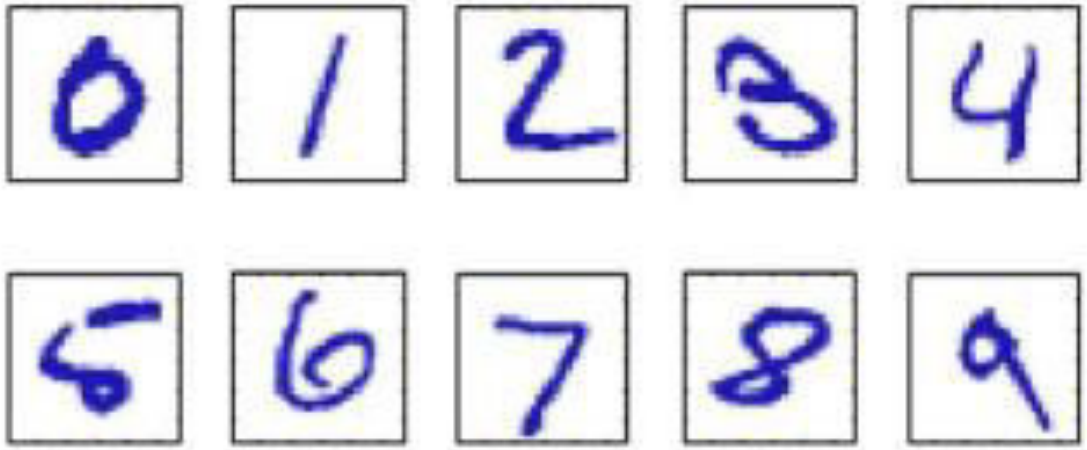
- Vision processing
- Language processing
- Pattern recognition
- Games and Expert systems
- Forecasting things like stock market trends, weather
- Data mining
- Fraud detection
- Financial modelling
- Medical diagnosis and Bioinformatics
- Robotics and Autonomous vehicles

Real-world Applications

- Spam email detection
- Netflix/YouTube recommendations
- Fraud detection in banking
- Medical diagnosis
- Self-driving cars

Handwritten Address Interpretation System

- One of first commercial and widely used ML systems (for zip codes & checks)



- Images are 28x28 pixels
- Represent input image as a vector $x \in \Re^{784}$
- Learn a classifier f(x) such that $f : x \rightarrow \{0,1,2,3,4,5,6,7,8,9\}$

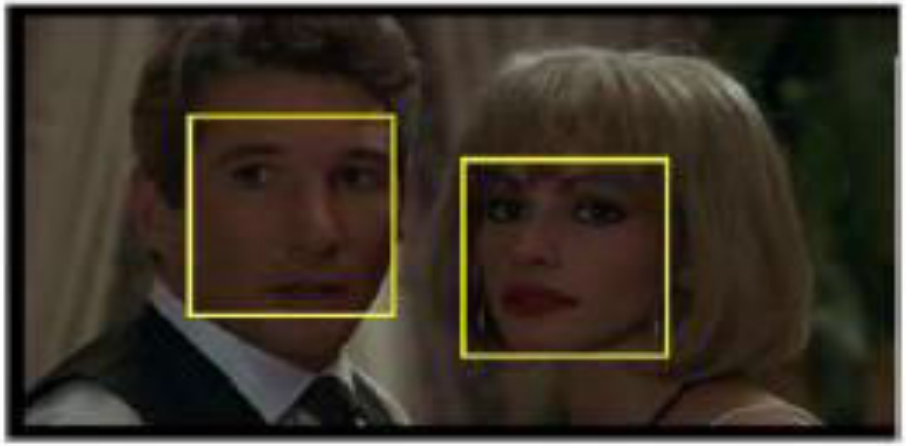
How to Proceed

- Collect some training samples for each digits
- Start learning and classifying
- Systems can achieve errors of 0.4%

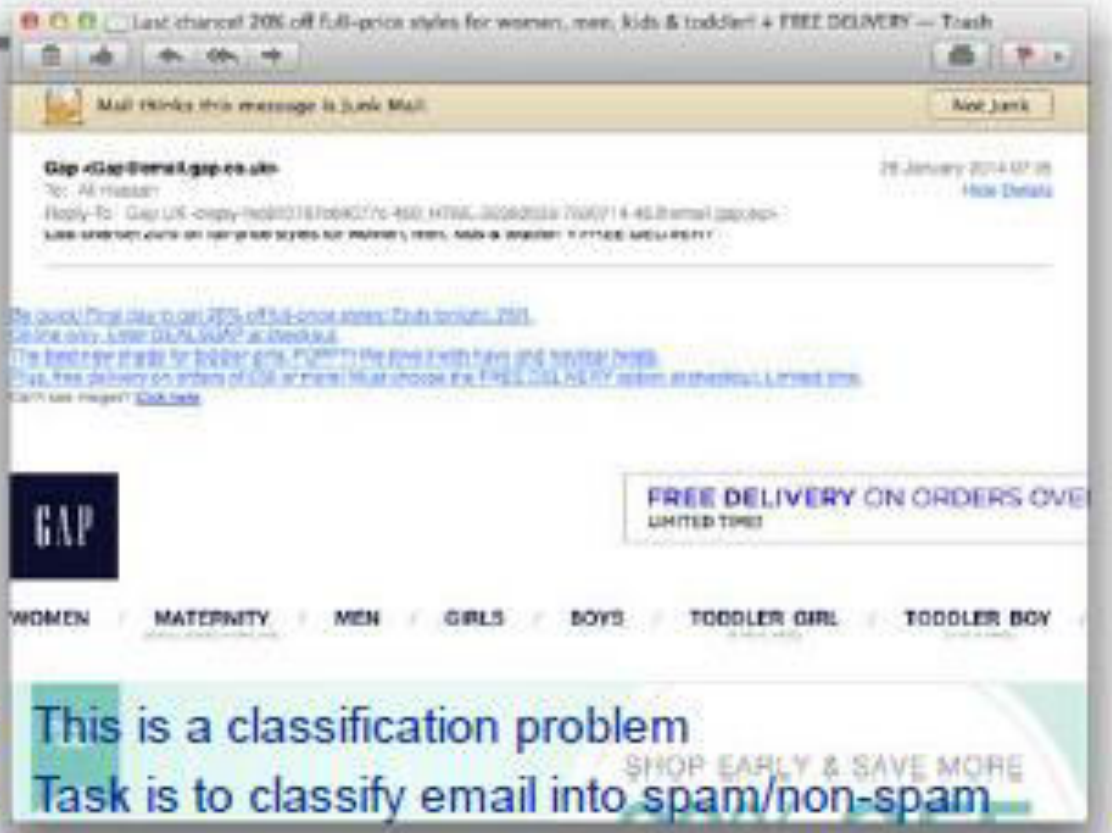


Example 2: Face Detection



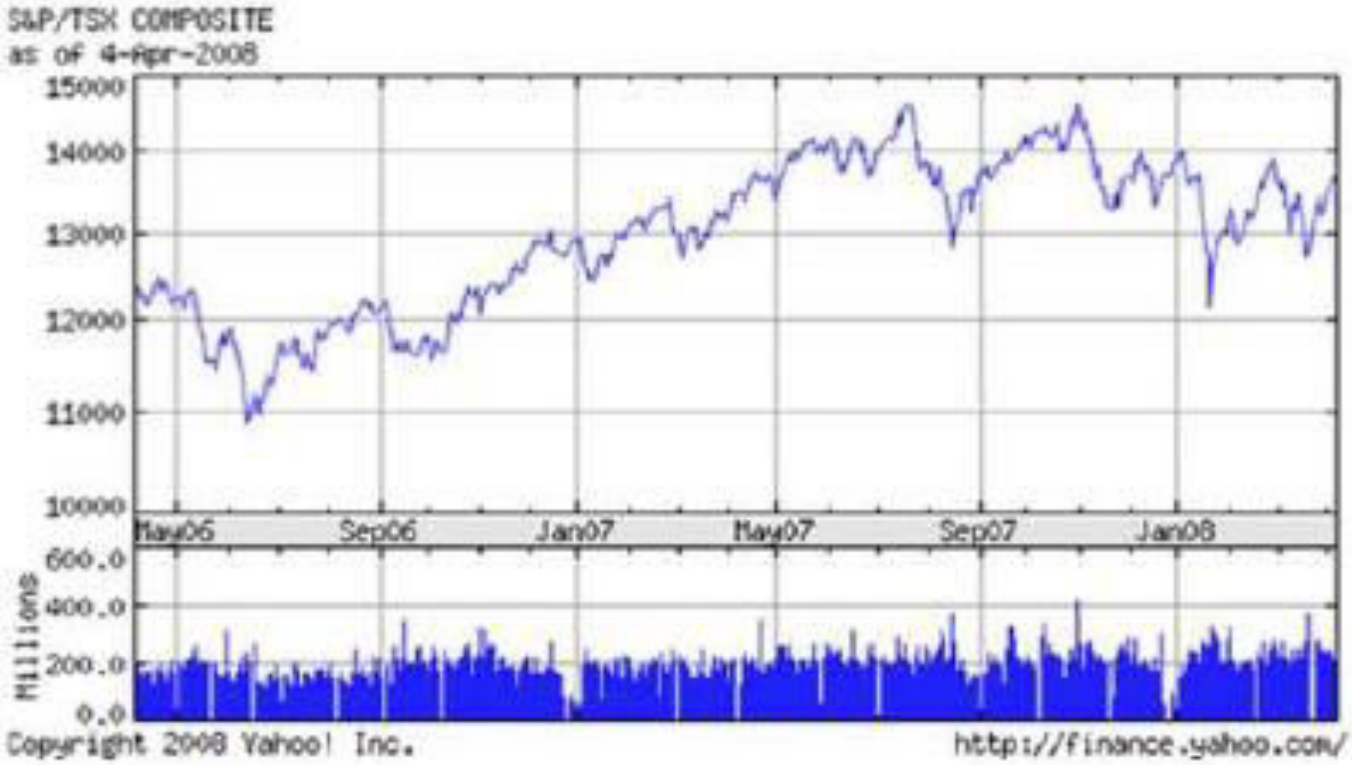


Spam Detection



- This is a classification problem
- Task is to classify email into spam/non-spam
- Data xi is word count, e.g. , outperform, "you may be surprised to be contacted" ...
- Requires a learning system as "enemy" keeps innovating

Stock Price Predictions



Language Translation



Recommender Systems



Basic Terminology

- **Dataset:** The entire collection of data used for the learning process.
- **Feature (x):** An individual measurable property or characteristic of the data (e.g., in house pricing, features would be square footage, number of rooms, or zip code).
- **Target/Label (y):** The specific output we are trying to predict (e.g., the actual price of the house).
- **Observation/Example:** A single row in the dataset containing a specific set of features and its corresponding label.
- **Model:** The specific mathematical function or "engine" that is produced after an algorithm has finished training on your data.

Training vs Testing Dataset

- **Training: learn patterns**
  - The observations in the training set form the experience that the algorithm uses to learn.
- **Testing: evaluate model**
  - The test set is a set of observations used to evaluate the performance of the model using some performance metric. It is important that no observations from the training set are included in the test set.
- **Avoid memorization**

Overfitting

- **Model memorizes data**
  - Memorizing the training set is called over-fitting. A program that memorizes its observations may not perform its task well, as it could memorize relations and structures that are noise or coincidence.
- **Poor performance on new data**
  - A program that generalizes well will be able to effectively perform a task with new data. In contrast, a program that memorizes the training data by learning an overly complex model could predict the values of the response variable for the training set accurately, but will fail to predict the value of the response variable for new examples.
- **High variance and poor generalization**
  - The model is too sensitive to the training data and changes a lot with small variations.

Underfitting

- Underfitting is when a machine learning model is too simple to learn the patterns in the data.
- Cannot capture relationships in data and makes large errors even on training data
- Balancing memorization and generalization, or over-fitting and under-fitting, is a problem common to many machine learning algorithms.

Validation Dataset

- In addition to the training and test data, a third set of observations, called a **validation** or **hold-out set**, is sometimes required.
- The validation set is used to tune variables called **hyper parameters**, which control how the model is learned.
- It is common to partition a single set of supervised observations into training, validation, and test sets. There are no requirements for the sizes of the partitions, and they may vary according to the amount of data available. It is common to allocate 50 percent or more of the data to the training set, 25 percent to the test set, and the remainder to the validation set.

Challenges in ML

- **Data quality issues**
  - Poor or noisy data → wrong predictions
  - Example: Missing or incorrect patient data → wrong diagnosis
- **Overfitting**
  - Model memorizes training data, fails on new data
  - Example: 100% training accuracy, but poor test performance
- **Computational cost**
  - ML models need high processing power and time
  - Example: Training deep learning models requires GPUs (Graphics Processing Unit)
- **Interpretability**
  - Hard to understand how complex models make decisions
  - Example: Neural network predicts loan rejection but gives no clear reason
  - Debugging issue: Difficult to find why model is wrong
  - Trust issue: Users don’t trust unclear decisions

Summary

ML learns from data

Three main types

Wide real-world applications